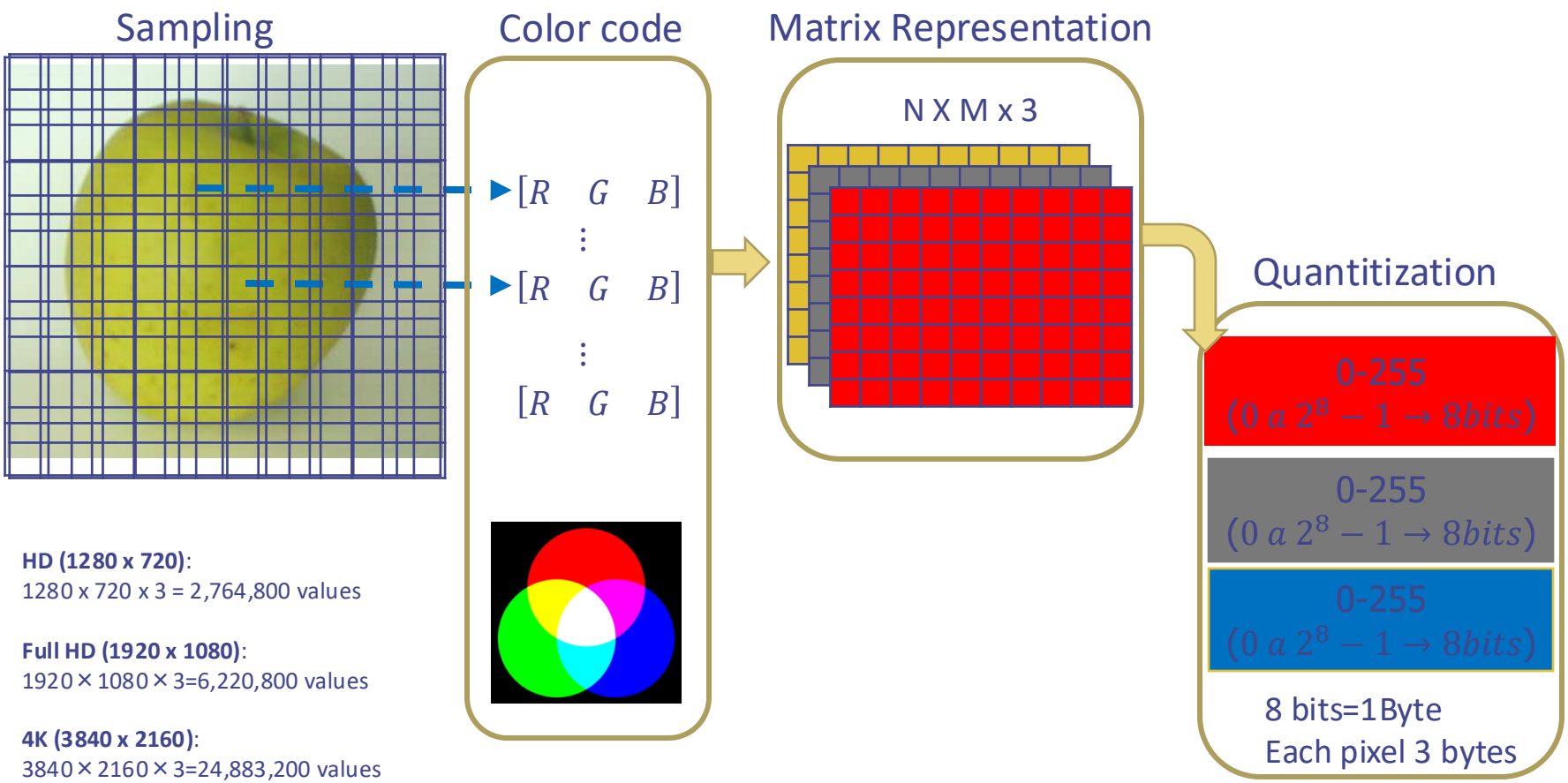
Engenharia de Características para Aprendizagem Computacional / Engenharia de Atributos

Application Domain: Image

Licenciatura em Enheraria Informática / Licenciatura em Inteligência Artificial e Ciência de Dados / Licenciatura em Engenharia e Ciência de Dados
Edition 2025-26

Marco Simões

Image Analysis



HD (1280 x 720):
1280 x 720 x 3 = 2,764,800 values

Full HD (1920 x 1080):
1920 × 1080 × 3=6,220,800 values

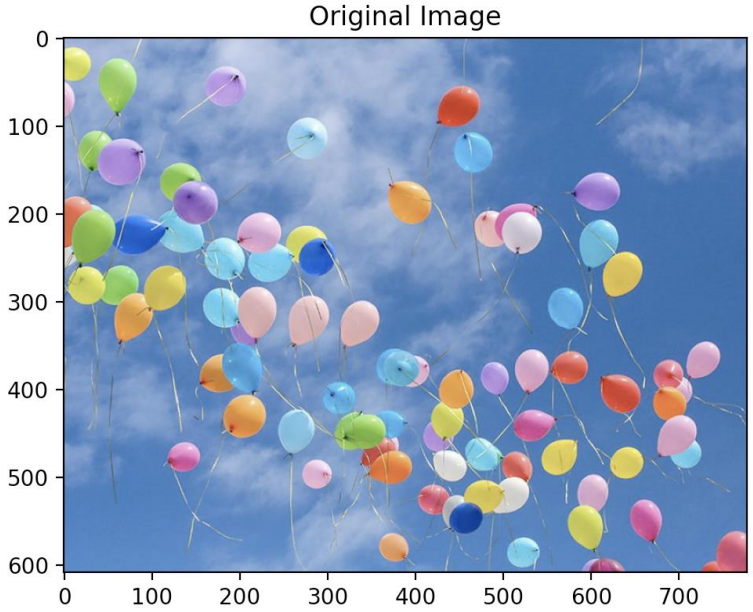
4K (3840 x 2160):
3840 × 2160 × 3=24,883,200 values

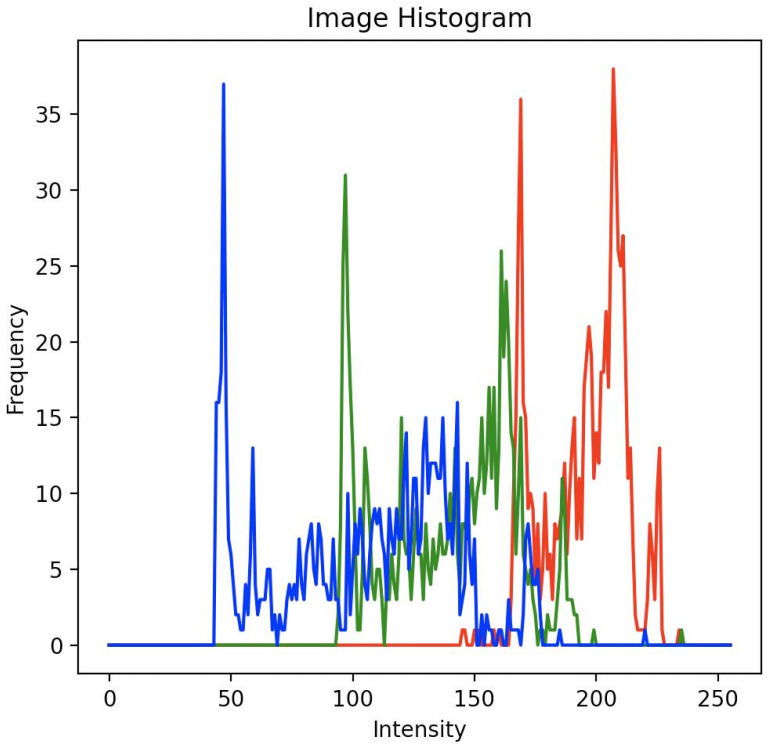
8K (7680 x 4320):
7680 × 4320 × 3=99,532,800 values

Low-level features

Color Histograms:

Number of pixels in the image with that color intensity



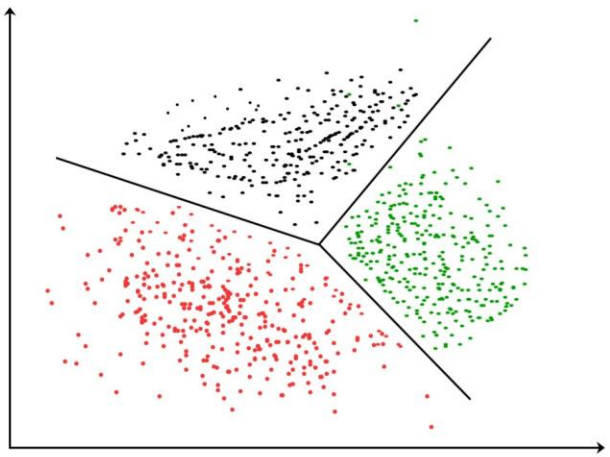


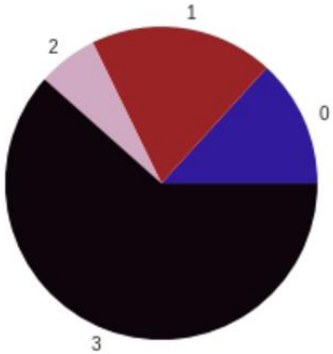
Low-level features

Dominant colors

What are the dominant colors of the picture?

- Apply K-means
- Centroids represent the dominant colors





python Copiar Editar

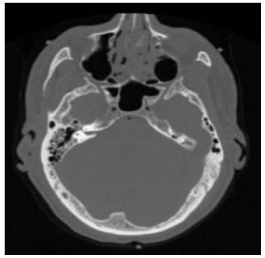
```python
from sklearn.cluster import KMeans

image_reshaped = image.reshape((-1, 3))  # Flatten image pixels
kmeans = KMeans(n_clusters=5)
kmeans.fit(image_reshaped)
dominant_colors = kmeans.cluster_centers_
print("Dominant Colors:", dominant_colors)
```

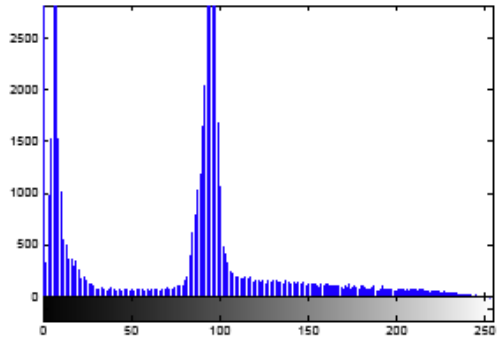
Entropy features

Color Entropy:

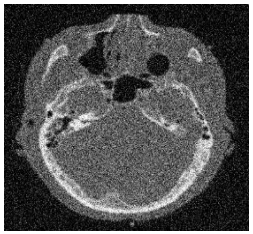
$$A \square \{a_1, a_2, \ldots, a_n\}$$

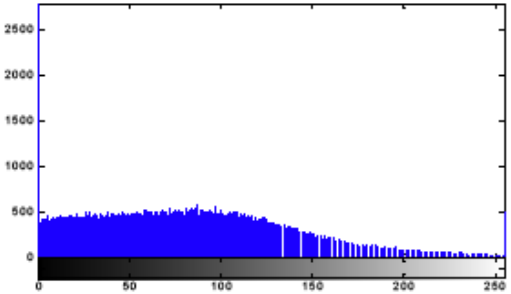
$$H(A) = -\square_{i=1}^{n} P(a_i) \log_2 P(a_i)$$





H=5.94 bits/p





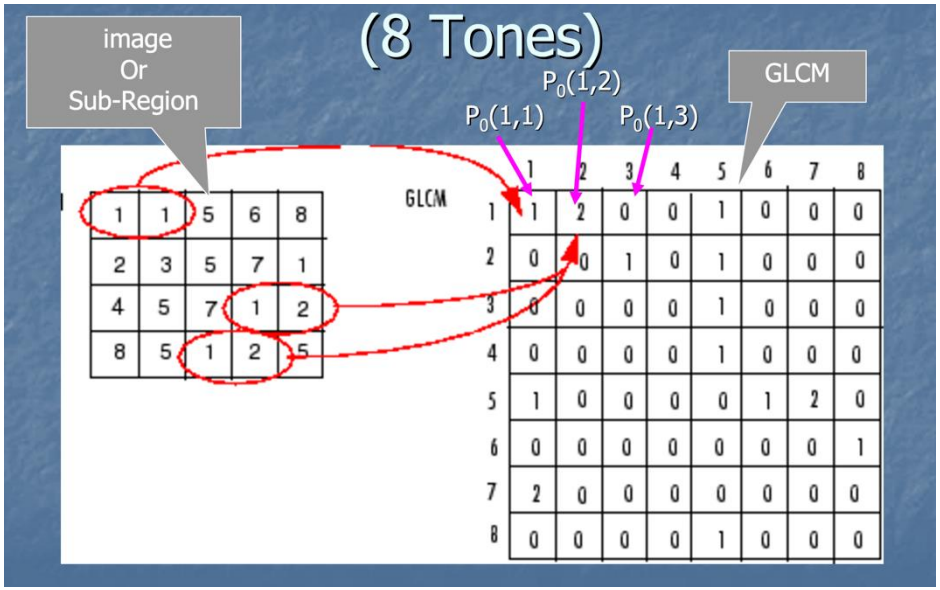
H=6.96 bits/p

Texture Features

Gray-Level Co-occurrence Matrix (GLCM)

- Calculated from the Gray-Scale version of the image (average the three color channels, for example)

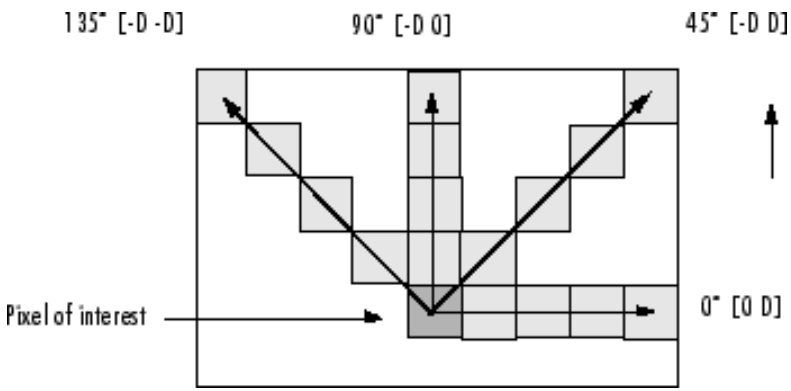
- Indicates how often two colors/intensities appear one after another in a predefined direction



Texture Features

Gray-Level Co-occurrence Matrix (GLCM)

- Directions: horizontal, vertical, diagonal, anti-diagonal



- Distances: How many pixels apart?

Texture Features

Gray-Level Co-occurrence Matrix (GLCM)

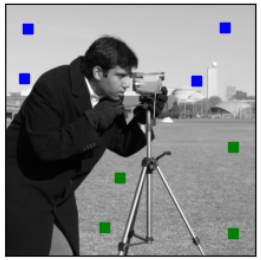
- From the resulting matrix, several features can be extracted:

No.	Texture measure	Formula
1	Mean (MEA)	$\mathrm{MEA} = \sum_{i,j=0}^{N-1} i(P_{i,j})$
2	Variance (VAR)	$\mathrm{VAR} = \sum_{i,j=0}^{N-1} P_{i,j}(i - \mathrm{MEA})^2$
3	Homogeneity (HOM)	$\mathrm{HOM} = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1+(i-j)^2}$
4	Contrast (CON)	$\mathrm{CON} = \sum_{i,j=0}^{N-1} P_{i,j}(i - j)^2$
5	Dissimilarity (DIS)	$\mathrm{DIS} = \sum_{i,j=0}^{N-1} P_{i,j}\lvert i - j\rvert$

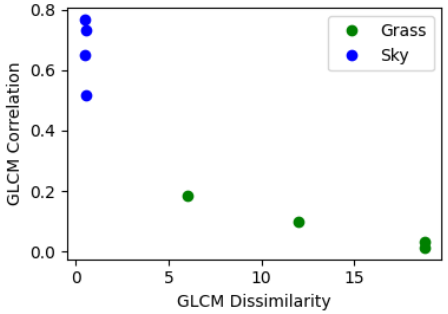
Texture Features

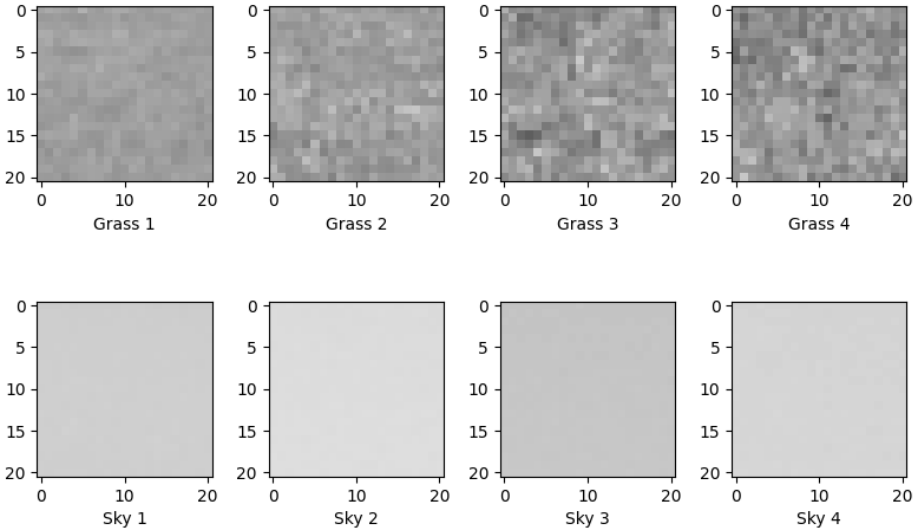
Gray-Level Co-occurrence Matrix (GLCM)

```
import skimage.feature as ft

gray_image = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
glcm = ft.greycomatrix(gray_image, distances=[1], angles=[0], levels=256
contrast = ft.greycoprops(glcm, 'contrast')[0, 0]
print("GLCM Contrast:", contrast)
```



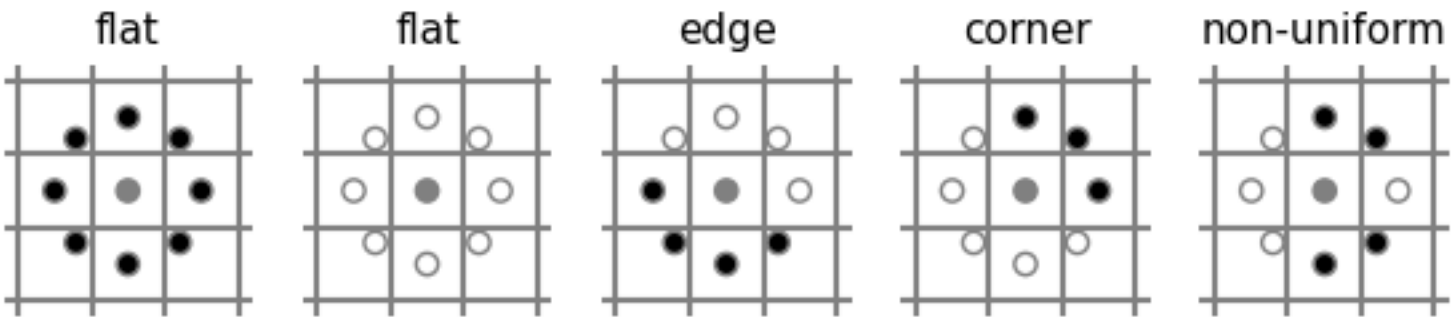
Original Image





Texture Features

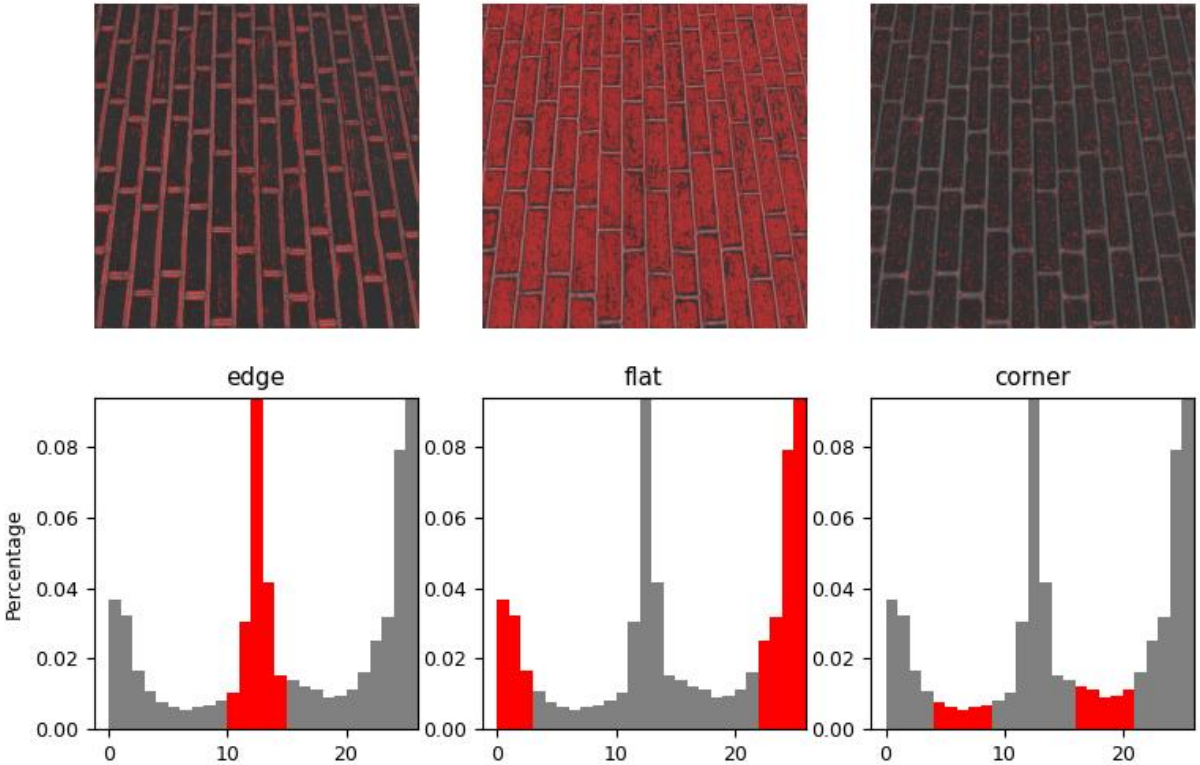
Local Binary Pattern (LBP)




```
image = data.brick()
lbp = local_binary_pattern(image, n_points, radius, METHOD)
```

Texture Features

Local Binary Pattern (LBP)



Histogram of Oriented Gradients (HOG)

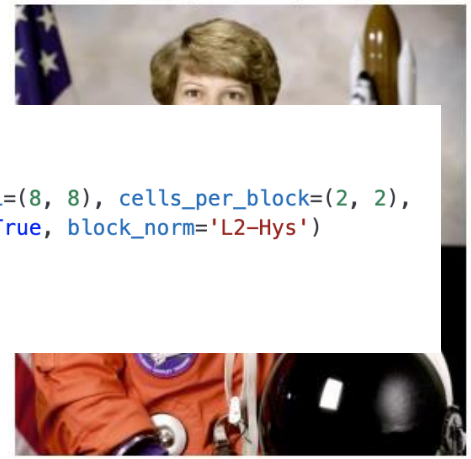
Algorithm overview

Compute a Histogram of Oriented Gradients (HOG) by

1. (optional) global image normalisation
2. computing the gradient image in x and y
3. computing gradient histograms
4. normalising across blocks
5. flattening into a feature vector

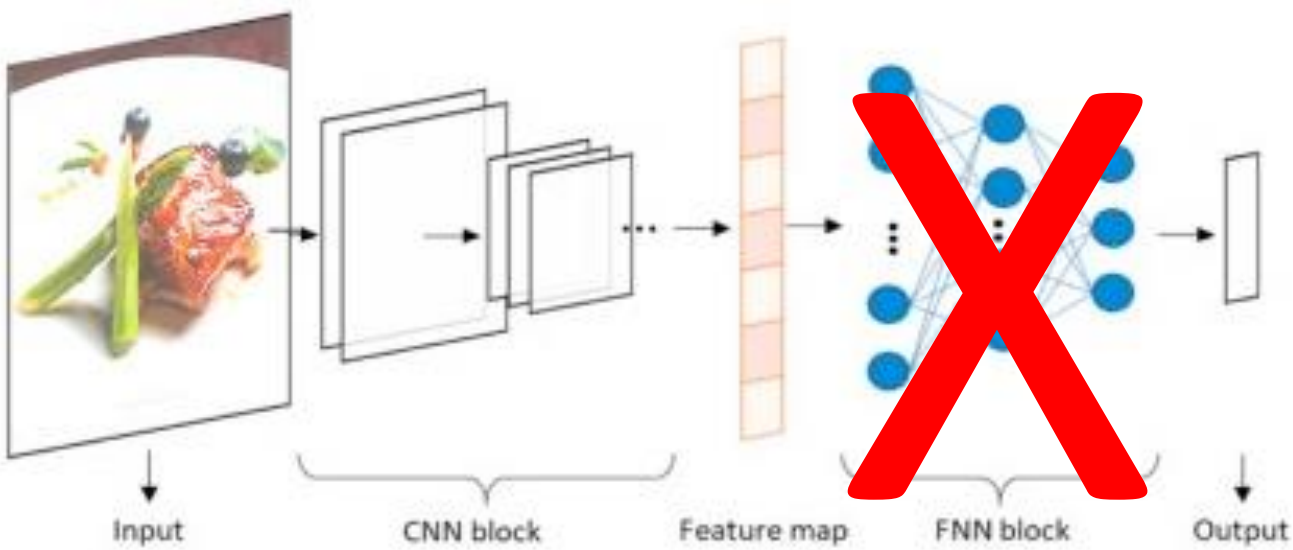
```
from skimage.feature import hog

hog_features, hog_image = hog(gray_image, pixels_per_cell=(8, 8), cells_per_block=(2, 2),
                              orientations=9, visualize=True, block_norm='L2-Hys')
plt.imshow(hog_image, cmap="gray")
plt.title("HOG Features")
plt.show()
```

Input image



Histogram of Oriented Gradients

Extract features from pre-trained CNNs



Extract features from pre-trained CNNs

```python
import torch
import torchvision.transforms as transforms
from torchvision.models import resnet50
from PIL import Image

# Load Pretrained ResNet50
model = resnet50(pretrained=True)
model = torch.nn.Sequential(*(list(model.children())[:-1]))  # Remove final classification layer
model.eval()

# Load Image
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor()
])
image = Image.open("image.jpg")
image = transform(image).unsqueeze(0)

# Extract Features
with torch.no_grad():
    features = model(image)

print("Extracted Features Shape:", features.shape)  # (1, 2048, 1, 1)
```

✅ **Alternative Pretrained Models**

- **VGG16 / VGG19**
- **EfficientNet**
- **Vision Transformers (ViT)**